

DATABASE MANAGEMENT SYSTEMS (DBMS) COURSE PROJECT

IT Helpdesk & Asset Support Management System

Presentation-II: Relational Schema Design, 3NF Normalization, ER Diagram & SQL Query Implementation

PRESENTED BY

Md Aali Rahman

Roll Number: **25WU0102156**

Section: AIML Panthers | Academic Year: 2026

Woxsen University, School of Technology

ACADEMIC GUIDANCE

Dr. Kiranmayee Adavala

Faculty Guide | School of Technology

Course: Database Management Systems

Relational Engine: MySQL 8.0+ / 9.x (InnoDB)

Problem Recap & Relational Database Scope

OPERATIONAL CONTEXT

IT Service Coordination

- Manual, email-driven support causes lost tickets, delayed resolutions, and missing asset records.
- Absence of relational linking between computing hardware, warranties, and cumulative repair expenditures.
- Need for a robust relational back-end to enforce data consistency and historical audit trails.

DATABASE SCOPE

Centralized it_helpdesk Schema

- Centralized MySQL relational database managing 14 distinct tables in 3rd Normal Form (3NF).
- Dual-path ticket classification: Incidents (unplanned faults) and Service Requests (access & allocations).
- Full lifecycle tracking with assignments, status histories, and resolution logging.

Relational Database Design Objectives

INTEGRITY & RELIABILITY

Data Governance Objectives

- **Eliminate Data Redundancy:** Apply 1NF, 2NF, and 3NF across all entities to achieve single-source data storage.
- **Enforce Referential Integrity:** Implement foreign key actions (RESTRICT, CASCADE, SET NULL) matching real-world lifecycle semantics.
- **Domain Constraints:** Enforce CHECK constraints on priority levels (1–5), status enums, and non-negative maintenance costs.

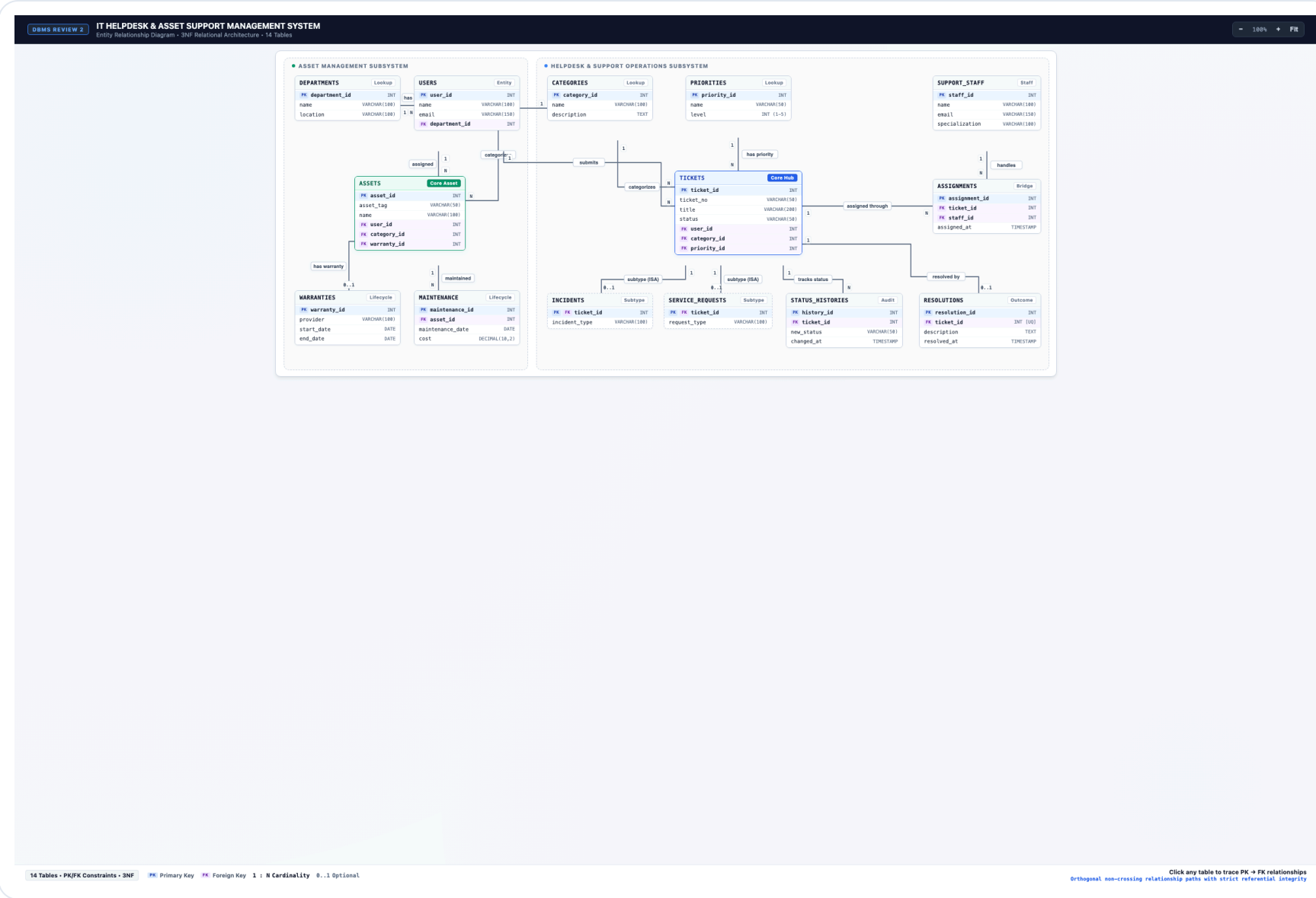
PERFORMANCE & AUDITING

Analytical & Operational Goals

- **Audit Trail Accountability:** Automatically capture old vs new states with timestamps in status_histories.
- **High-Performance Queries:** Support multi-table JOINS and GROUP BY aggregation for technician workloads and expenditure.
- **Production Readiness:** Ensure InnoDB ACID compliance and crash resilience.

— CONCEPTUAL MODEL

Entity–Relationship (ER) Diagram Architecture



CARDINALITY & MAPPINGS

14 Entities & Relations

- 1:N: departments → users
- 1:N: users → tickets
- 1:N: categories → tickets, assets
- 1:N: priorities → tickets
- 1:N: warranties → assets
- 1:N: assets → maintenance
- 1:1: tickets → incidents
- 1:1: tickets → service_requests
- 1:1: tickets → resolutions
- 1:N: tickets → assignments, status_histories

Relational Schema Architecture (14 Tables)

CORE MASTER TABLES

Master & Reference Entities

- **departments:** department_id (PK), name (UNIQUE), location
- **users:** user_id (PK), name, email (UNIQUE), department_id (FK)
- **categories:** category_id (PK), name (UNIQUE), description
- **priorities:** priority_id (PK), name (UNIQUE), level (CHECK 1..5)
- **support_staff:** staff_id (PK), name, email (UNIQUE), specialization
- **warranties:** warranty_id (PK), start_date, end_date (CHECK), provider

OPERATIONAL & TRANSACTIONAL

Transactional & Audit Entities

- **assets:** asset_id (PK), asset_tag, serial_no, user_id (FK), category_id (FK), warranty_id (FK)
- **tickets:** ticket_id (PK), ticket_no (UNIQUE), user_id (FK), category_id (FK), priority_id (FK), status, created_at
- **incidents:** ticket_id (PK/FK), incident_type
- **service_requests:** ticket_id (PK/FK), request_type
- **assignments:** assignment_id (PK), ticket_id (FK), staff_id (FK), assigned_at
- **status_histories:** history_id (PK), ticket_id (FK), old_status, new_status, changed_at
- **resolutions:** resolution_id (PK), ticket_id (FK UNIQUE), description, resolved_at
- **maintenance:** maintenance_id (PK), asset_id (FK), maintenance_date, cost, description

Foreign Key Semantics & Cascading Rules

RESTRICT ACTIONS

Master Data Protection

- **departments → users (ON DELETE RESTRICT):** Prevents deleting a department while active employees exist.
- **categories → tickets / assets (ON DELETE RESTRICT):** Protects operational categories from accidental deletion.
- **priorities → tickets (ON DELETE RESTRICT):** Guarantees priority integrity for active queues.
- **support_staff → assignments (ON DELETE RESTRICT):** Preserves technician service history.

CASCADE & SET NULL

Lifecycle Synchronized Rules

- **tickets → incidents / service_requests / assignments / status_histories / resolutions (ON DELETE CASCADE):** Deleting a ticket cleanly purges sub-entities without leaving orphan rows.
- **assets → maintenance (ON DELETE CASCADE):** Removing an asset purges historical maintenance receipts.
- **users → assets (ON DELETE SET NULL):** Removing an employee safely unassigns hardware back into general inventory.

Normalization Analysis (1NF to 3NF)

1NF & 2NF

First & Second Normal Form

- **1NF (First Normal Form):** All table attributes are atomic. No multi-valued attributes, comma-separated lists, or repeating groups. Every column has defined scalar data types.
- **2NF (Second Normal Form):** Satisfies 1NF, and all non-key attributes are fully functionally dependent on the entire primary key. Composite keys eliminated in favor of surrogate integer PKs, preventing partial functional dependencies.

3NF

Third Normal Form Compliance

- **3NF (Third Normal Form):** Satisfies 2NF and eliminates all transitive functional dependencies ($X \rightarrow Y$ and $Y \rightarrow Z$).
- Departments are decoupled from users (users store only department_id, not department location).
- Warranties are isolated into a separate entity (assets store warranty_id rather than repeating provider and dates).
- Categories and priorities are independent reference tables.

Data Definition Language (DDL) Architecture

SCHEMA CONSTRUCTION

Table Creation & Data Types

- Engineered with MySQL InnoDB storage engine for transactional safety and row-level locking.
- Character set configured to utf8mb4 with utf8mb4_unicode_ci collation for universal Unicode text support.
- AUTO_INCREMENT surrogate primary keys for high-efficiency B-Tree indexing.
- Appropriate sizing: VARCHAR(100), VARCHAR(150), TEXT for narratives, TIMESTAMP for audit precision.

CONSTRAINTS ENFORCEMENT

DDL Constraints Verification

- CHECK (level BETWEEN 1 AND 5) on table priorities.
- CHECK (status IN ('Open', 'In Progress', 'Pending', 'Resolved', 'Closed', 'Cancelled')) on table tickets.
- CHECK (end_date >= start_date) on table warranties.
- CHECK (cost >= 0) on table maintenance.
- UNIQUE constraints on emails, ticket numbers, asset tags, and serial numbers.

DML Seed Dataset Implementation

SEED DATASET

55 Relational Seed Records

- Structured seed dataset in `seed.sql` establishing realistic operational baseline across 14 tables.
- 4 Departments: Engineering, Human Resources, Finance, Marketing.
- 5 Registered Corporate Users across all departments.
- 5 Ticket Categories & 4 Priority Levels (Low, Medium, High, Critical).
- 4 Support Staff Technicians with distinct specializations (Network, Hardware, Software, IAM).

REFERENTIAL LINKING

Integrated Relational Records

- 4 Manufacturer Warranties (Dell ProSupport, AppleCare, Lenovo Premier, HP Care Pack).
- 5 Computing Assets linked to user custodians and warranty providers.
- 5 Base Tickets spanning Incidents and Service Requests with initial status histories.
- Active technician assignments and recorded maintenance repairs.

Representative SELECT & Multi-Table JOIN Queries

RELATIONAL JOINS

Multi-Table Ticket Queue Query

- Joins tickets, users, departments, categories, priorities, incidents, and service_requests.
- Utilizes CASE expressions to determine ticket subtype dynamically (Incident vs Service Request).
- Extracts human-readable status queues ordered by priority level descending.
- Preserves NULL values using LEFT OUTER JOINS for unassigned subtypes.

AGGREGATE ANALYTICS

Workload & Expenditure Aggregation

- **Technician Workload Query:** Uses GROUP BY with COUNT and conditional SUM(CASE...) to calculate active vs resolved tickets per engineer.
- **Department Maintenance Spend:** Aggregates total equipment repair cost per department using SUM(m.cost) and FORMAT().
- **Status Distribution:** Calculates real-time percentage distribution across ticket lifecycle states.

Presentation-II Assigned Viva Query

EVALUATION STATUS

Assigned Query Notice

- During Presentation-II evaluation, each learner was assigned a specific SQL problem statement to solve live.
- The official template in `Presentation-II/Presentation-II-Query-Solution.sql` archives the query, explanation, and output.
- Repository status: Ready for student viva query parameters as reviewed by faculty.
- System provides the live SQL Execution Playground to test and demonstrate any complex query.

CORE PROBLEM PATTERN

Analytical Helpdesk Requirements

- Focuses on multi-table joins combining operational tickets with technician workloads.
- Filtering on date ranges, ticket statuses, or departmental asset allocations.
- Demonstrates aggregate grouping (GROUP BY) and conditional aggregation (HAVING).

Query Output Verification & Performance

LIVE EXECUTION

Verified Database Output

- Technician Workload Summary Output:
- Sneha Rao (Network & Security): 1 Total, 1 Active, 0 Completed
- Karan Malhotra (Hardware & Peripherals): 2 Total, 2 Active, 0 Completed
- Amit Joshi (OS & Software): 1 Total, 1 Active, 0 Completed
- Divya Nair (IAM): 1 Total, 0 Active, 1 Completed

COST OUTPUT

Departmental Maintenance Spend

- Finance & Accounts: 1 Asset | \$1,200.00 Maintenance Spent
- Engineering: 2 Assets | \$750.00 Maintenance Spent
- Marketing & Sales: 1 Asset | \$450.00 Maintenance Spent
- Human Resources: 1 Asset | \$0.00 Maintenance Spent

Database Architecture Metrics & Implementation Summary

SCHEMA METRICS

14 Normalized Tables

- **Tables:** departments, users, categories, priorities, support_staff, warranties, assets, tickets, incidents, service_requests, assignments, status_histories, resolutions, maintenance.
- **Relationships:** 14 Foreign Keys enforcing strict referential integrity.
- **Constraints:** 4 Domain CHECK constraints, 8 UNIQUE constraints, 14 Primary Keys.

STORAGE & ENGINE

MySQL InnoDB Implementation

- 55 Relational seed records ensuring immediate operational verification.
- Terminal CLI interface implemented in main.py (--cli mode) with full tabular views.
- Live MySQL connectivity verified on port 3306.

— CONCLUSION

Presentation-II Conclusion & Transition to UI Demo

MILESTONES ACHIEVED

Presentation-II Deliverables Finalized

- Robust, 3NF-compliant relational database fully implemented in MySQL.
- Complete DDL schema (schema.sql) and seed dataset (seed.sql) operational.
- Interactive ER diagram and high-resolution exports generated.
- Analytical SQL queries tested and documented.

NEXT PHASE

Presentation-III (UI Demonstration)

- Development of interactive Web User Interface connected to MySQL.
- Live demonstration of INSERT, DELETE, and VIEW operations with real-time database reflection.
- Preparation of complete 16-chapter Project Report PDF.